PADHAKU
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CHAPTER 5

Prime Time

NCERT Exercise Solutions

5.1 Common Multiples and Common Factors · Idli-Vada Game	1 Q
Figure it Out · 5.1 (Idli-Vada)	8 Q
Jump Jackpot	4 Q
Figure it Out · 5.1	14 Q
5.2 Prime Numbers	5 Q
Figure it Out · 5.2	16 Q
5.3 Co-prime numbers for safekeeping treasures	5 Q
5.4 Prime Factorisation	3 Q
Figure it Out · 5.4 (Prime factorisation)	8 Q
Using prime factorisation to check co-primes	1 Q
Figure it Out · 5.4 (Co-prime and divisibility)	10 Q
5.5 Divisibility Tests	9 Q

✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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5.1 Common Multiples and Common Factors · Idli-Vada Game

? Numbers that are multiples of both 3 and 5 (15, 30, 45, ...) are called _____.

✔ ANSWER

Common multiples of 3 and 5.

Figure it Out · 5.1 (Idli-Vada)

1 At what number is 'idli-vada' said for the 10th time?

✔ ANSWER

'Idli-vada' is said at common multiples of 3 and 5: 15, 30, 45, ...

10th one = 10 × 15

150

2 (a) Game played for 1 to 90: how many times is 'idli' said (including 'idli-vada')?

✔ ANSWER

Multiples of 3 up to 90: 90 ÷ 3 = 30

30 times

2 (b) How many times is 'vada' said (including 'idli-vada')?

✔ ANSWER

Multiples of 5 up to 90: 90 ÷ 5 = 18

18 times

2 (c) **How many times is ‘idli-vada’ said?**

✔ ANSWER

Multiples of 15 up to 90: 90 ÷ 15 = 6

6 times

3 **What if the game was played till 900? How would your answers change?**

✔ ANSWER

Each answer becomes 10 times bigger.

‘Idli’: 900 ÷ 3 = **300 times**

‘Vada’: 900 ÷ 5 = **180 times**

‘Idli-vada’: 900 ÷ 15 = **60 times**

4 **Is Fig. 5.1 (two overlapping circles: multiples of 3, multiples of 5, common part) related to the ‘idli-vada’ game? Draw it for the game till 60.**

✔ ANSWER

Yes. Left circle = ‘idli’ numbers, right circle = ‘vada’ numbers, the overlap = ‘idli-vada’ numbers.

Till 60:

- Only multiples of 3: 3, 6, 9, 12, 18, 21, 24, 27, 33, 36, 39, 42, 48, 51, 54, 57

- Only multiples of 5: 5, 10, 20, 25, 35, 40, 50, 55
- Common (middle): **15, 30, 45, 60**

? **Play the game with (a) 2 and 5, (b) 3 and 7, (c) 4 and 6, up to 60. Which numbers go in the common part of the figure?**

✔ ANSWER

(a) 2 and 5: common multiples **10, 20, 30, 40, 50, 60**

(b) 3 and 7: common multiples **21, 42**

(c) 4 and 6: common multiples **12, 24, 36, 48, 60**

The other multiples go in the two outer parts of each figure.

? **Two numbers were used; one was 4. Nobody said just 'vada' (only 'idli' or 'idli-vada'). Which could be the other number: 2, 3, 5, 8, 10?**

✔ ANSWER

'Vada' is for the larger number. Nobody says just 'vada' only if every multiple of the larger number is also a multiple of the smaller one.

- If the other number is 8: every multiple of 8 is a multiple of 4. ✓
- If it is 2: 4 is larger; every multiple of 4 is a multiple of 2. ✓

- 3, 5, 10 do not work (e.g. 3: 3 itself would be ‘idli’, 4 ‘vada’).

8 (2 also works).

⭐ Jump Jackpot

? **What jump sizes can reach both 15 and 30? Find them all.**

✔ ANSWER

Factors of 15: 1, 3, 5, 15

Factors of 30: 1, 2, 3, 5, 6, 10, 15, 30

Common factors: **1, 3, 5, 15**

1 **In the table (31–70), is there anything common among the shaded numbers?**

✔ ANSWER

They are all multiples of 3.

2 **Is there anything common among the circled numbers?**

✔ ANSWER

They are all multiples of 4.

3 **Which numbers are both shaded and circled? What are these numbers called?**

✔ ANSWER

36, 48, 60

They are **common multiples of 3 and 4** (multiples of 12).

Figure it Out · 5.1

1 **Find all multiples of 40 that lie between 310 and 410.**

✔ ANSWER

40 × 8 = 320, 40 × 9 = 360, 40 × 10 = 400

320, 360, 400

2 (a) **Who am I? A number less than 40. One of my factors is 7. The sum of my digits is 8.**

✔ ANSWER

Multiples of 7 below 40: 7, 14, 21, 28, 35

Digit sums: 7, 5, 3, 10, 8

35

2 (b) **Who am I? A number less than 100. Two of my factors are 3 and 5. One of my digits is 1 more than the other.**

✔ ANSWER

Multiples of 15 below 100: 15, 30, 45, 60, 75, 90

Only 45 has digits differing by 1 (5 = 4 + 1).

45

3 **Find a perfect number between 1 and 10 (sum of all factors = twice the number).**

✔ ANSWER

Factors of 6: 1, 2, 3, 6

Sum = 12 = 2 × 6

6

4 (a) **Common factors of 20 and 28.**

✔ ANSWER

20: 1, 2, 4, 5, 10, 20

28: 1, 2, 4, 7, 14, 28

1, 2, 4

4 (b) **Common factors of 35 and 50.**

✔ ANSWER

35: 1, 5, 7, 35

50: 1, 2, 5, 10, 25, 50

1, 5

4 (c) **Common factors of 4, 8 and 12.**

✔ ANSWER

4: 1, 2, 4

8: 1, 2, 4, 8

12: 1, 2, 3, 4, 6, 12

1, 2, 4

4 (d) **Common factors of 5, 15 and 25.**

✔ ANSWER

5: 1, 5

15: 1, 3, 5, 15

25: 1, 5, 25

1, 5

5 **Find any three numbers that are multiples of 25 but not multiples of 50.**

✔ ANSWER

Odd multiples of 25 work.

25, 75, 125 (also 175, 225, ...)

6 **'Idli-vada' game with two numbers both smaller than 10. The first 'idli-vada' is said after 50. What could the two numbers be?**

✔ ANSWER

The first 'idli-vada' is the smallest common multiple; it must be more than 50.

- 7 and 8 → 56
- 7 and 9 → 63
- 8 and 9 → 72

7 and 8, 7 and 9, or 8 and 9

7 **Treasures on 28 and 70. What jump sizes land on both?**

✔ ANSWER

Factors of 28: 1, 2, 4, 7, 14, 28

Factors of 70: 1, 2, 5, 7, 10, 14, 35, 70

Common factors: **1, 2, 7, 14**

8 **Guna erased all numbers except the common multiples 24, 48, 72. What could the two numbers be? Fill the empty regions.**

💡 SAMPLE ANSWER

The first common multiple is 24, so the pair could be **6 and 8** (or 3 and 8).

For 6 and 8 (up to 72):

- Only multiples of 6: 6, 12, 18, 30, 36, 42, 54, 60, 66
- Only multiples of 8: 8, 16, 32, 40, 56, 64
- Common: 24, 48, 72

(check the picture in your book)

9 **Find the smallest number that is a multiple of all numbers from 1 to 10, except 7.**

✔ ANSWER

It must be a multiple of 8, 9 and 5 (these cover 2, 3, 4, 6, 10 too).

$8 \times 9 \times 5 = 360$

360

10 **Find the smallest number that is a multiple of all numbers from 1 to 10.**

✔ ANSWER

Now 7 is also needed.

360 × 7 = 2520

2520

⭐ 5.2 Prime Numbers

1 **In how many ways can Guna arrange 12 figs in a rectangle?**

✔ ANSWER

1 × 12, 2 × 6, 3 × 4, 4 × 3, 6 × 2, 12 × 1

Rows and columns are factors of 12.

6 arrangements (3 different shapes, each turned).

2 **In how many ways can Anshu arrange 7 figs in a rectangle?**

✔ ANSWER

Only 1 × 7 or 7 × 1.

Only one arrangement (7 has only two factors, 1 and 7).

? **How many prime numbers are there from 21 to 30? How many composite numbers?**

✔ ANSWER

Primes: 23, 29 → **2 primes**

Composites: 21, 22, 24, 25, 26, 27, 28, 30 → **8 composites**

? **Can we list all the prime numbers from 1 to 100? (Sieve of Eratosthenes)**

✔ ANSWER

Yes. The 25 primes are:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

? **Think how the Sieve method works.**

💡 SAMPLE ANSWER

Each time we circle a number, it was not crossed out, so no smaller number divides it: it is prime.

Then we cross out its multiples, because they have it as a factor: they are composite.

So at the end only primes stay circled.

Figure it Out · 5.2

1 **2 is a prime and an even number. Is there any other even prime?**

✔ ANSWER

Every other even number has 2 as a factor, so it has more than two factors.

No. 2 is the only even prime.

2 **In the primes till 100, what is the smallest and the largest difference between two successive primes?**

✔ ANSWER

Smallest: 3 − 2 = **1**

Largest: 97 − 89 = **8**

3 **Are there an equal number of primes in every row (decade) of the table? Which decades have the least and the most primes?**

✔ ANSWER

Decade	Primes	Count
1–10	2, 3, 5, 7	4
11–20	11, 13, 17, 19	4
21–30	23, 29	2
31–40	31, 37	2
41–50	41, 43, 47	3

51–60	53, 59	2
61–70	61, 67	2
71–80	71, 73, 79	3
81–90	83, 89	2
91–100	97	1

No. Least: **91–100** (1 prime). Most: **1–10 and 11–20** (4 each).

4 **Which of these are prime: 23, 51, 37, 26?**

✔ ANSWER

51 = 3 × 17 and 26 = 2 × 13 are composite.

23 and 37

5 **Write three pairs of prime numbers less than 20 whose sum is a multiple of 5.**

✔ ANSWER

2 + 3 = 5

3 + 7 = 10

2 + 13 = 15

(2, 3), (3, 7), (2, 13) (others: (3, 17), (7, 13), (11, 19))

6 **13 and 31 are primes with the same digits. Find such pairs of primes up to 100.**

✔ ANSWER

17 and 71, 37 and 73, 79 and 97 (besides 13 and 31)

7 **Find seven consecutive composite numbers between 1 and 100.**

✔ ANSWER

90, 91, 92, 93, 94, 95, 96

(91 = 7 × 13, 93 = 3 × 31; the others are even or end in 5.)

8 **Twin primes differ by 2 (e.g. 3 and 5, 17 and 19). Find the other twin primes between 1 and 100.**

✔ ANSWER

5 and 7, 11 and 13, 29 and 31, 41 and 43, 59 and 61, 71 and 73

9 (a) **True or false? There is no prime number whose units digit is 4.**

✔ ANSWER

True. A number ending in 4 is even and bigger than 2, so 2 is a factor; it is composite.

9 (b) **True or false? A product of primes can also be prime.**

✔ ANSWER

False. e.g. 2 × 3 = 6 has factors 2 and 3 besides 1 and 6, so it is composite.

9 (c) **True or false? Prime numbers do not have any factors.**

✔ ANSWER

False. A prime has exactly two factors: 1 and itself.

9 (d) **True or false? All even numbers are composite numbers.**

✔ ANSWER

False. 2 is even but it is prime.

9 (e) **True or false? 2 and 3 are both primes; for every other prime, the next number is composite.**

✔ ANSWER

True. Every prime after 2 is odd, so the next number is even and bigger than 2, hence composite.

10 **Which is the product of exactly three distinct primes: 45, 60, 91, 105, 330?**

✔ ANSWER

45 = 3 × 3 × 5; 60 = 2 × 2 × 3 × 5; 91 = 7 × 13

105 = 3 × 5 × 7; 330 = 2 × 3 × 5 × 11

105

11 **How many three-digit primes can you make using each of 2, 4 and 5 once?**

✔ ANSWER

Numbers: 245, 254, 425, 452, 524, 542

Each ends in 2, 4 (even) or 5 (multiple of 5).

None (0)

12 **3 is prime and 2 × 3 + 1 = 7 is prime. Find at least five other primes for which doubling and adding 1 gives a prime.**

✔ ANSWER

2 × 2 + 1 = 5

2 × 5 + 1 = 11

2 × 11 + 1 = 23

2 × 23 + 1 = 47

2 × 29 + 1 = 59

2, 5, 11, 23, 29 (also 41 → 83)

5.3 Co-prime numbers for safekeeping treasures

a–d **Jump size 1 is not allowed. Check if these pairs are safe: (a) 15 and 39 (b) 4 and 15 (c) 18 and 29 (d) 20 and 55**

✔ ANSWER

(a) 15 and 39: jump 3 reaches both → **not safe**

(b) 4 and 15: no common factor except 1 → **safe**

(c) 18 and 29: no common factor except 1 → **safe**

(d) 20 and 55: jump 5 reaches both → **not safe**

a–e **Which pairs are co-prime? (a) 18 and 35 (b) 15 and 37 (c) 30 and 415 (d) 17 and 69 (e) 81 and 18**

✔ ANSWER

(a) 18 = 2 × 3 × 3, 35 = 5 × 7 → **co-prime**

(b) 15 = 3 × 5, 37 is prime → **co-prime**

(c) both have factor 5 → not co-prime

(d) 17 is prime, 69 = 3 × 23 → **co-prime**

(e) both have factor 9 → not co-prime

1, 2 **Anshu: (1) sometimes the first common multiple equals the product of the two numbers; (2) sometimes it is less. Find examples. How is it related to being co-prime?**

✔ ANSWER

(1) 3 and 5 → 15 = 3 × 5; 4 and 9 → 36 = 4 × 9

(2) 4 and 6 → 12, but 4 × 6 = 24; 6 and 15 → 30, but 6 × 15 = 90

If the two numbers are co-prime, the first common multiple is their product. If not, it is less than the product.

? **Thread art: in some diagrams the thread touches every peg, in some it does not. Is it related to the number of pegs and the thread-gap being co-prime?**

💡 SAMPLE ANSWER

Yes. 12 pegs, gap 4 (not co-prime): only 3 pegs are used.

13 pegs, gap 3 (co-prime): every peg is used.

When pegs and gap are co-prime, the thread touches every peg.

a–d **Make thread pictures for: (a) 15 pegs, gap 10 (b) 10 pegs, gap 7 (c) 14 pegs, gap 6 (d) 8 pegs, gap 3**

✔ ANSWER

(a) 15 and 10 share 5 → touches only 15 ÷ 5 = **3 pegs**

(b) 10 and 7 co-prime → touches **all 10 pegs**

(c) 14 and 6 share 2 → touches only 14 ÷ 2 = **7 pegs**

(d) 8 and 3 co-prime → touches **all 8 pegs**

⭐ 5.4 Prime Factorisation

? Anshu wrote 56 = 14 × 4 and 63 = 21 × 3 and said 56 and 63 are co-prime. Where did Anshu go wrong?

✔ ANSWER

One way of writing a number as a product does not show **all** its factors.

56 = 7 × 8 and 63 = 7 × 9 show that **7 is a common factor**, so they are not co-prime.

? Using the diagram, explain why 30 = 2 × 3 × 5, no matter which way you multiply 2, 3 and 5.

✔ ANSWER

(2 × 3) × 5 = 6 × 5 = 30

(3 × 5) × 2 = 15 × 2 = 30

(2 × 5) × 3 = 10 × 3 = 30

The order of multiplying does not change the product.

? **72 = 12 × 6, 12 = 2 × 2 × 3, 6 = 2 × 3. What is the prime factorisation of 72? Compare how often each prime occurs.**

✔ ANSWER

72 = 2 × 2 × 3 × 2 × 3 = **2 × 2 × 2 × 3 × 3**

2 occurs 3 times (two from 12 + one from 6).

3 occurs 2 times (one from 12 + one from 6).

Figure it Out · 5.4 (Prime factorisation)

1 **Find the prime factorisations: 64, 104, 105, 243, 320, 141, 1728, 729, 1024, 1331, 1000.**

✔ ANSWER

64 = **2 × 2 × 2 × 2 × 2 × 2**

104 = **2 × 2 × 2 × 13**

105 = **3 × 5 × 7**

243 = **3 × 3 × 3 × 3 × 3**

320 = **2 × 2 × 2 × 2 × 2 × 2 × 5**

141 = **3 × 47**

1728 = **2 × 2 × 2 × 2 × 2 × 2 × 3 × 3 × 3**

729 = **3 × 3 × 3 × 3 × 3 × 3**

1024 = **2 × 2 × 2 × 2 × 2 × 2 × 2 × 2 × 2 × 2**

1331 = **11 × 11 × 11**

1000 = **2 × 2 × 2 × 5 × 5 × 5**

2 **A number's prime factorisation has one 2, two 3s and one 11. What is the number?**

✔ ANSWER

$2 \times 3 \times 3 \times 11 = 18 \times 11$

198

3 **Find three primes, all less than 30, whose product is 1955.**

✔ ANSWER

1955 ends in 5: $1955 \div 5 = 391$

$391 = 17 \times 23$

$5 \times 17 \times 23 = 1955$

4 (a) **Prime factorisation of 56×25 without multiplying first.**

✔ ANSWER

$56 = 2 \times 2 \times 2 \times 7$, $25 = 5 \times 5$

$2 \times 2 \times 2 \times 5 \times 5 \times 7$

4 (b) **Prime factorisation of 108×75 without multiplying first.**

✔ ANSWER

$108 = 2 \times 2 \times 3 \times 3 \times 3$, $75 = 3 \times 5 \times 5$

$2 \times 2 \times 3 \times 3 \times 3 \times 3 \times 5 \times 5$

4 (c) **Prime factorisation of 1000 × 81 without multiplying first.**

✔ ANSWER

1000 = 2 × 2 × 2 × 5 × 5 × 5, 81 = 3 × 3 × 3 × 3

2 × 2 × 2 × 3 × 3 × 3 × 3 × 5 × 5 × 5

5 (a) **Smallest number whose prime factorisation has three different primes?**

✔ ANSWER

Use the three smallest primes: 2 × 3 × 5

30

5 (b) **Smallest number whose prime factorisation has four different primes?**

✔ ANSWER

2 × 3 × 5 × 7

210

⭐ Using prime factorisation to check co-primes

? **80 = 2 × 2 × 2 × 2 × 5 and 63 = 3 × 3 × 7 have no common prime factor. If they had a common composite factor, would its prime factors appear in both factorisations?**

✔ ANSWER

Yes. Any prime factor of a common factor would divide both numbers, so it would appear in both factorisations.

Since no prime is common, 80 and 63 have no common factor except 1: they are **co-prime**.

Figure it Out · 5.4 (Co-prime and divisibility)

1 (a) Are 30 and 45 co-prime?

✔ ANSWER

30 = 2 × 3 × 5, 45 = 3 × 3 × 5

Common primes 3 and 5 → **not co-prime**

1 (b) Are 57 and 85 co-prime?

✔ ANSWER

57 = 3 × 19, 85 = 5 × 17

No common prime → **co-prime**

1 (c) Are 121 and 1331 co-prime?

✔ ANSWER

121 = 11 × 11, 1331 = 11 × 11 × 11

Common prime 11 → **not co-prime**

1 (d) Are 343 and 216 co-prime?

✔ ANSWER

343 = 7 × 7 × 7, 216 = 2 × 2 × 2 × 3 × 3 × 3

No common prime → **co-prime**

2 (a) **Is 225 divisible by 27?**

✔ ANSWER

225 = 3 × 3 × 5 × 5, 27 = 3 × 3 × 3

27 needs three 3s, 225 has only two → **No**

2 (b) **Is 96 divisible by 24?**

✔ ANSWER

96 = 2 × 2 × 2 × 2 × 2 × 3, 24 = 2 × 2 × 2 × 3

24's factorisation is included in 96's → **Yes** (96 = 24 × 4)

2 (c) **Is 343 divisible by 17?**

✔ ANSWER

343 = 7 × 7 × 7, 17 is prime

17 does not appear in 343 → **No**

2 (d) **Is 999 divisible by 99?**

✔ ANSWER

999 = 3 × 3 × 3 × 37, 99 = 3 × 3 × 11

11 does not appear in 999 → **No**

3 **First number = 2 × 3 × 7, second = 3 × 7 × 11. Are they co-prime? Does one divide the**

other?

✔ ANSWER

Common primes 3 and 7 → **not co-prime**

2 is not in the second; 11 is not in the first → **neither divides the other**

(42 and 231)

4 **Guna says, "Any two prime numbers are co-prime." Is he right?**

✔ ANSWER

Yes (for two different primes). Each prime has only 1 and itself as factors, so the only common factor is 1.

e.g. 2 and 3, 5 and 11.

⭐ 5.5 Divisibility Tests

? **Does 8560 have any factors from 2 to 10? Which can be checked without long division?**

✔ ANSWER

Easy checks: 2, 4, 5, 8, 10 (look at the last digits).

8560 ends in 0 → divisible by 2, 5, 10; last two digits 60 → by 4; last three 560 → by 8.

8560 is divisible by 2, 4, 5, 8 and 10.

? **Divisibility by 10: Is 125 a multiple of 10? Is 8560 divisible by 10? Do you agree that numbers divisible by 10 end with '0'?**

✔ ANSWER

Multiples of 10 all end in 0.

125 ends in 5 → **not a multiple of 10**; it will not appear in the list.

8560 ends in 0 → **Yes**.

Agree.

? **Divisibility by 5: What is the pattern? What is the largest number less than 399 divisible by 5? Is 8560 divisible by 5? Do you agree multiples of 5 end in 0 or 5?**

✔ ANSWER

Last digits of multiples of 5 go 5, 0, 5, 0, ...

Largest below 399: **395**

8560 ends in 0 → **Yes**

Agree.

? **Divisibility by 2: Is 682 divisible by 2? Is 8560? Do you agree multiples of 2 end in 0, 2, 4, 6 or 8? Multiples of 2 between 399 and 411?**

✔ ANSWER

682 ends in 2 → **Yes**. 8560 ends in 0 → **Yes**.

Agree.

Between 399 and 411: **400, 402, 404, 406, 408, 410**

? **Divisibility by 4: Numbers between 330 and 340, 1730 and 1740, 2030 and 2040 divisible by 4? What do you observe? Is 8536 divisible by 4?**

✔ ANSWER

330–340: **332, 336**

1730–1740: **1732, 1736**

2030–2040: **2032, 2036**

They always end in 32 and 36: only the last two digits matter.

8536: 36 is divisible by 4 → **Yes**

1–3 **Do you agree? (1) Only the last two digits matter for divisibility by 4. (2) If the last two digits form a multiple of 4, the number is divisible by 4. (3) If the number is divisible by 4, its last two digits form a multiple of 4.**

✔ ANSWER

Agree with all three.

Why: 100 is divisible by 4, so the hundreds and above part (e.g. 8500 in 8536) is always a multiple of 4. Only the last two digits decide.

? **Divisibility by 8: Can the last two digits be used? Numbers between 120 and 140, 1120**

and 1140, 3120 and 3140 divisible by 8? What do you observe?

✔ ANSWER

120–140: **128, 136**

1120–1140: **1128, 1136**

3120–3140: **3128, 3136**

The last two digits are not enough (128 is divisible by 8, 228 is not). The **last three digits** decide.

? **Change the last two digits of 8560 so that the result is a multiple of 8.**

✔ ANSWER

8500 ÷ 8 leaves remainder 4, so add 4, 12, 20, ...

8504, 8512, 8520, 8528, 8536, 8544, 8552 (any one; 8568, 8576 ... also work)

1–3 **Do you agree? (1) Only the last three digits matter for divisibility by 8. (2) If the last three digits form a multiple of 8, the number is divisible by 8. (3) If the number is divisible by 8, the last three digits form a multiple of 8.**

✔ ANSWER

Agree with all three.

Why: 1000 = 8 × 125, so the thousands and above part is always a multiple of 8. Only the last three digits decide.

Figure it Out · 5.5

1 (a) **From the year you were born till now, which years were leap years?**

💡 SAMPLE ANSWER

Example for a student born in 2014: **2016, 2020, 2024**.

(Use your own birth year; list multiples of 4.)

1 (b) **From 2024 till 2099, how many leap years are there?**

✔ ANSWER

2024, 2028, ..., 2096 (2100 is not counted)

(2096 − 2024) ÷ 4 + 1 = 18 + 1

19 leap years

2 **Find the largest and smallest 4-digit palindromes divisible by 4.**

✔ ANSWER

A palindrome abba ends in ‘ba’; the last digit a must be even (it is also the first digit).

Largest: a = 8, need ‘b8’ divisible by 4 → b = 8 → **8888**

Smallest: a = 2, need ‘b2’ divisible by 4 → b = 1 (12) → **2112**

3 (a) **Always, sometimes or never true: sum of two even numbers is a multiple of 4.**

✔ ANSWER

2 + 6 = 8 ✓, but 2 + 4 = 6 ×

Sometimes true

3 (b) **Always, sometimes or never true: sum of two odd numbers is a multiple of 4.**

✔ ANSWER

1 + 3 = 4 ✓, but 1 + 5 = 6 ×

Sometimes true

4 **Find remainders when 78, 99, 173, 572, 980, 1111, 2345 are divided by (a) 10, (b) 5, (c) 2.**

✔ ANSWER

Number	÷ 10	÷ 5	÷ 2
78	8	3	0
99	9	4	1
173	3	3	1
572	2	2	0
980	0	0	0
1111	1	1	1

2345	5	0	1

5 **Guna checked 14560 for divisibility by only two of 2, 4, 5, 8, 10 and declared it divisible by all. Which two?**

✔ ANSWER

Divisible by 8 → also by 4 and 2.

Divisible by 8 (even) and 5 → also by 10.

8 and 5 (8 and 10 also works)

6 **Which are divisible by all of 2, 4, 5, 8 and 10: 572, 2352, 5600, 6000, 77622160?**

✔ ANSWER

Must end in 0 and last three digits divisible by 8.

572, 2352 do not end in 0 → no.

5600 (600 ÷ 8 = 75), 6000 (000), 77622160 (160 ÷ 8 = 20) → yes

5600, 6000, 77622160

7 **Write two numbers whose product is 10000, neither ending in 0.**

✔ ANSWER

10000 = 2 × 2 × 2 × 2 × 5 × 5 × 5 × 5

Keep all 2s together and all 5s together.

16 × 625 = 10000

⭐ 5.6 Fun with numbers · Special numbers

? **Box: 9, 16, 25, 43. Which number looks special? Why?**

💡 SAMPLE ANSWER

- 9: only single-digit number; only multiple of 3
- 16: only even number
- 25: only multiple of 5
- 43: only prime; only number that is not a square

? **Box: 5, 7, 12, 35. Say how each number is special.**

💡 SAMPLE ANSWER

- 5: smallest number
- 7: largest single-digit number
- 12: only even number
- 35: only multiple of both 5 and 7 ($35 = 5 \times 7$)

? **Box: 3, 8, 11, 24. Say how each number is special.**

💡 SAMPLE ANSWER

- 3: only odd single-digit number
- 8: only single-digit even number (a cube, $2 \times 2 \times 2$)
- 11: only 2-digit prime

- 24: only 2-digit even number (24 = 3 × 8)

? **Box: 27, 3, 123, 31. Say how each number is special.**

💡 SAMPLE ANSWER

- 3: only single-digit number
- 27: only cube (3 × 3 × 3)
- 123: only 3-digit number
- 31: only number that is not a multiple of 3

? **Box: 17, 27, 44, 65. Say how each number is special.**

💡 SAMPLE ANSWER

- 17: only prime
- 27: only multiple of 3
- 44: only even number
- 65: only multiple of 5

⭐ A prime puzzle

1 **Fill the grid with primes so each row's product is on the right and each column's product is below.**
Rows: 105, 20, 30. Columns: 28, 125, 18.

✔ ANSWER

Column 125 = 5 × 5 × 5, so the middle column is all 5s.

Col 1	Col 2	Col 3	Row product
7	5	3	105
2	5	2	20
2	5	3	30

Columns: 7 × 2 × 2 = 28, 5 × 5 × 5 = 125, 3 × 2 × 3 = 18 ✓

2 **Rows: 8, 105, 70. Columns: 30, 70, 28.**

✔ ANSWER

Row 8 = 2 × 2 × 2.

Col 1	Col 2	Col 3	Row product
2	2	2	8
3	5	7	105
5	7	2	70

Columns: 2 × 3 × 5 = 30, 2 × 5 × 7 = 70, 2 × 7 × 2 = 28 ✓

3 **Rows: 63, 27, 190. Columns: 45, 42, 171.**

✔ ANSWER

Row 27 = 3 × 3 × 3; 19 (from 190) goes in column 171.

Col 1	Col 2	Col 3	Row product
3	7	3	63
3	3	3	27
5	2	19	190

Columns: 3 × 3 × 5 = 45, 7 × 3 × 2 = 42, 3 × 3 × 19 = 171 ✓

4 **Rows: 343, 66, 44. Columns: 28, 154, 231.**

✔ ANSWER

Row 343 = 7 × 7 × 7.

Col 1	Col 2	Col 3	Row product
7	7	7	343
2	11	3	66
2	2	11	44

Columns: 7 × 2 × 2 = 28, 7 × 11 × 2 = 154, 7 × 3 × 11 = 231 ✓

🎉 **Chapter 5 done! Revise the answers once more tomorrow.**